Prompt

Prompt 1:

I used the inbuilt ai it got me the codes and ran it itself I didn't do anything i just want to confirm if this answers any questions that can mr make of the data set given.

Prompt 2:

You are a senior data scientist working on a credit risk platform case study.

I have multiple CSV files from a lending platform:

1. loan_core.csv

   Columns: id, term, installment, grade, loan_status, disbursement_method 2. borrower_profile.csv

   (7 borrower-related columns such as income, employment, etc.)

3. credit_history.csv

4. account_balances.csv

5. account_activity.csv

6. extra_unassigned.csv

The objective of the case is:

To evaluate whether the platform's loan grading system (Grade A– G) still correctly reflects borrower risk, and whether there are signs of drift, inconsistency, or hidden irregularities.

Tasks for you:

1. Load all CSV files using pandas.
2. Inspect row counts and confirm whether datasets are rowaligned or require merging. 3. Define a binary default_flag using:

- Charged Off
- Does not meet credit policy – Charged Off
- Default

4. Compute:

- Overall default rate
- Default rate by grade
- Default rate by term
- Default rate by disbursement_method

5. Combine borrower_profile with loan_core and analyze:

- Income distribution by grade
- Default rate by income bands
- Employment length vs default
- Any mismatch between grade and borrower strength

6. Use credit_history to analyze:

- Whether credit metrics align with grade
- Whether similar credit profiles receive different grades

7. Perform drift analysis:
- If any time variable exists, compute default rate by grade over time
- Identify instability or grade compression
8. Identify anomalies:
- Similar borrowers with different outcomes
- High-income borrowers defaulting at high rate
- Low-risk grade showing elevated default
9. Produce:
- Clear summary tables
- Key statistical findings
- Insights formatted as presentation-ready bullet points
- Code organized cleanly with comments

The goal is not just to compute metrics, but to critically evaluate whether the grading system is stable, predictive, and internally consistent.

Write clean, efficient, scalable pandas code.

Prompt 3:

You are a senior quantitative risk analyst reviewing a lending platform's grading system.

I already completed descriptive analysis showing:

- Overall default rate = 11.61%
- Grade A–G default rates are monotonic (3.18% to 37.43%)
- Evidence of grade compression over time
- Income and credit metrics analyzed descriptively

Now I need ADVANCED VALIDATION beyond descriptive groupby tables.

Perform the following analyses using clean, well-commented Python code:

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1) PREDICTIVE POWER VALIDATION

- Run a logistic regression model using ONLY grade as predictor of default_flag.
- Compute:
- AUC (ROC)

- Accuracy
- Confusion matrix
- Evaluate whether grade alone is a strong predictor.

Then:

- Build a second model using key borrower + credit features.
- Compare AUC of:

  Model 1: Grade only

  Model 2: Full feature set
- Determine whether grade captures most predictive information or is redundant.

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2) ECONOMIC RISK VALIDATION

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interest rate or loan amount is available:

- Estimate expected loss = PD × LGD × EAD
- Approximate LGD if not available (use reasonable assumption, state clearly)
- Compare expected loss across grades.
- Determine whether risk increase from A→G is proportional or nonlinear.

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3) GRADE DRIFT STATISTICAL TEST

- Quantify whether grade compression over time is statistically significant.
- Compute year-wise AUC for grade.
- Test whether predictive power declines over time.

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4) MISALIGNMENT TEST

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Control for income and key credit variables.

- Within similar borrower profiles:
- Test whether grade still significantly predicts default.
- Identify if grading system adds incremental predictive value.

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5) ACTIONABLE CONCLUSIONS

Generate:

- Clear bullet-point executive conclusions.
- Whether model retraining is recommended.
- Whether recalibration is necessary.
- Whether pricing adjustments are required.
- Whether anomalies suggest data quality or underwriting issues.

Output format:

- Clean summary tables

- Statistical metrics clearly labeled
- Executive-ready conclusions (no fluff)
- Highlight risks and recommended actions

Prompt 4:

Make a read me files of all the outcomes I have sent.

Prompt 5:

For all sections in my python code make a chart for each sections and send me the codes with instructions.